

computational power and storage powered by semiconductors has been astonishing. Today, we can run very large language models (LLMs) with billions of parameters in seconds on our mobile phones—a task that would have taken a significant supercomputer decades ago.

This quantum leap in semiconductor chips and storage has made AI successful, deployable, and accessible to various applications.

The growth in semiconductors and electronics has been the silent but indispensable driving force behind the success of AI, communication, electric vehicles, and other domains. What was once considered a computational challenge beyond reach is now a reality due to the strides made in these technological fields. Today's computing power on mobile devices dwarfs what was available just a few years ago, demonstrating the monumental progress witnessed. These technologies have transformed daily lives and propelled AI and other applications to new heights, enabling it to harness its potential in previously unimaginable ways. The intertwined evolution of electronics, semiconductors, and AI showcases the power of human innovation and our ability to shape the future through technology.

Electronics: A thriving sector and manufacturing surge

India has seen remarkable growth and transformation in recent years. It is not just a nation with a burgeoning population of young and aspirational individuals but also a rapidly expanding market for electronics. The journey of India's economic progress has been fascinating, and the evolution of its electronics industry is a pivotal part of this narrative.

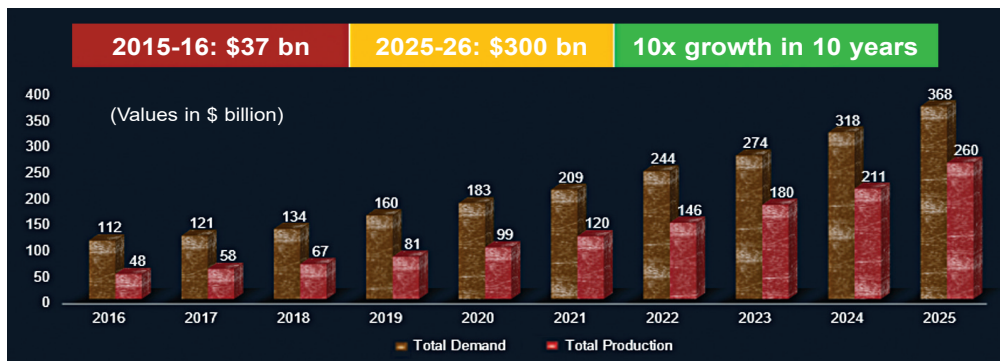


Fig. 1: India's growth in the manufacturing sector

India's gross domestic product (GDP) reached the milestone of one trillion dollars (\$1T) after nearly 67 years of independence. However, the subsequent years have witnessed an accelerated pace of economic growth. It took just seven years to reach the two-trillion-dollar mark (\$2T) and another three years to accomplish the third trillion (\$3T).

This growth reflects the dynamic nature of India's economy. The electronics sector has been on a remarkable trajectory at the heart of this growth. The consumption of electronics in India is surging from \$67 billion in 2015 to over \$310 billion in 2026, making it one of the country's fastest-growing sectors. This sector's expansion is driven by the aspirations of the Indian population and their increasing appetite for electronic devices for various aspects of life.

Along with the market, India's manufacturing capabilities in the electronics sector have grown significantly over the past decade. From a meagre \$35-40 billion in 2015-16, the nation is now poised to reach the \$300 billion mark for electronics manufacturing before the decade's end with more than \$100 billion in exports. India has done an excellent job creating large-scale manufacturing capabilities that are growing quickly.

While India may manufacture a substantial portion of electronics for domestic consumption and exports, global companies design and own

many of these products. More than 90% of the manufacturing in India caters to global brands and companies, leaving domestic product design and ownership at less than 10%. This situation poses several challenges. First, when India does not design its products, it has limited control over the choice of components used in these devices. Foreign entities primarily make decisions about components, suppliers, and quality in their headquarters, limiting India's influence in semiconductor purchases.

For instance, consider the iPhone. While it may be manufactured in India, the crucial decisions regarding its components, such as processors, memory, and connectivity chips, are made by Apple in Cupertino and its suppliers in places like Shenzhen, China. On the economic side, manufacturing companies, including global giants like Foxconn and Indian manufacturing companies, earn a small fraction of the product's value, less than 10%. At the same time, the product and brand owner reap significant economic value and profits of upwards of 30%.

India needs to convert its outstanding 'M'arket potential, strong 'M'anufacturing foundation, and abundance of quality 'M'anpower to develop India-designed products and brands to harness the economic and strategic potential of the electronics sector. This shift will enable the country to retain more value in production, create jobs, and enhance